

Hyperbolic functions



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Hyperbolic functions



Introduction: Hyperbolic functions

- ▶ In this section we define the **hyperbolic sine**, the **hyperbolic cosine** and the **hyperbolic tangent**, built from the exponential function e^x
- ▶ We study their parity, sign, monotonicity and graphs
- ▶ These functions appear often in calculus (limits, derivatives, integrals) and have properties very similar to those of the trigonometric functions

Hyperbolic sine and cosine

Definition 1: hyperbolic sine function and hyperbolic cosine function

The hyperbolic functions:

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto \operatorname{Sh} x = \frac{e^x - e^{-x}}{2}$$

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto \operatorname{Ch} x = \frac{e^x + e^{-x}}{2}$$

*are called **hyperbolic sine** and **hyperbolic cosine**.*

Parity and fundamental relations

- ▶ The hyperbolic sine is an **odd** function, while the hyperbolic cosine is an **even** function. Hence we have:

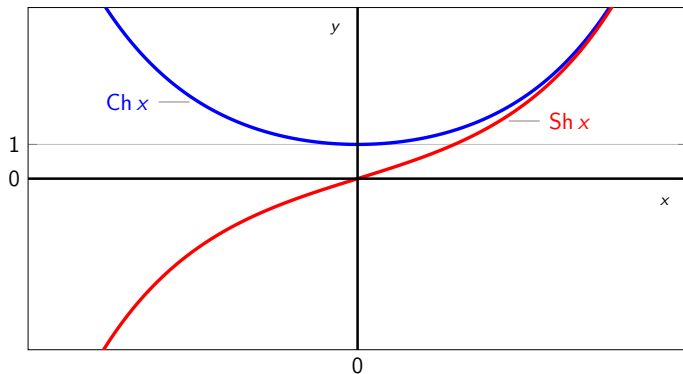
$$\operatorname{Sh}(-x) = -\operatorname{Sh} x, \quad \forall x \in \mathbb{R} \quad \text{and} \quad \operatorname{Ch}(-x) = \operatorname{Ch} x, \quad \forall x \in \mathbb{R}$$

- ▶ We have the fundamental relations:

$$\operatorname{Ch}^2 x - \operatorname{Sh}^2 x = 1, \quad \forall x \in \mathbb{R}$$

$$\operatorname{Sh} x \leq \frac{e^x}{2} \leq \operatorname{Ch} x, \quad \forall x \in \mathbb{R}$$

Graphs of $\operatorname{Sh} x$ and $\operatorname{Ch} x$



Sign and monotonicity

- For the sign and monotonicity of the hyperbolic sine function we have:

$$\operatorname{Sh} x = 0 \iff x = 0 \quad \left\{ \begin{array}{ll} \operatorname{Sh} x > 0 & \text{if } x > 0 \\ \operatorname{Sh} x < 0 & \text{if } x < 0 \end{array} \right.$$

$$f \text{ is increasing} \quad \forall x \in \mathbb{R}$$

- For the sign and monotonicity of the hyperbolic cosine function we have:

$$\operatorname{Ch} x > 0, \quad \forall x \in \mathbb{R}$$

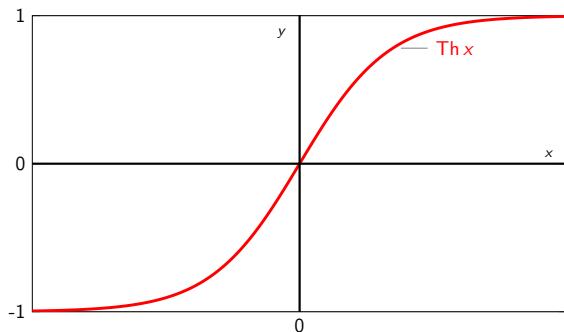
$$\left\{ \begin{array}{ll} \text{if } x < 0 & f \text{ is decreasing} \\ \text{if } x > 0 & f \text{ is increasing} \end{array} \right.$$

Hyperbolic tangent

The hyperbolic function:

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto \operatorname{Th} x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

is called the **hyperbolic tangent** and it is an odd function.



Hyperbolic functions:

- ▶ $\text{Sh } x = \frac{e^x - e^{-x}}{2}$ is **odd**, increasing on $\mathbb{R}$ and vanishes only at $x = 0$
- ▶ $\text{Ch } x = \frac{e^x + e^{-x}}{2}$ is **even**, always positive, decreasing for $x < 0$ and increasing for $x > 0$
- ▶ $\text{Ch}^2 x - \text{Sh}^2 x = 1$ and $\text{Sh } x \leq \frac{e^x}{2} \leq \text{Ch } x$ for every $x \in \mathbb{R}$
- ▶ $\text{Th } x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ is **odd**, with values in $(-1, 1)$

Main hyperbolic formulas



Introduction: Main hyperbolic formulas

- ▶ In this section we collect the **addition** and **double-argument** formulas for the hyperbolic functions
- ▶ They are the analogue of the trigonometric formulas (with some sign differences) and follow directly from the definitions through the exponential
- ▶ They are useful to simplify expressions and in computations with limits, derivatives and integrals

Addition

$$\operatorname{Sh}(x_1 + x_2) = \operatorname{Sh} x_1 \operatorname{Ch} x_2 + \operatorname{Sh} x_2 \operatorname{Ch} x_1$$

$$\operatorname{Ch}(x_1 + x_2) = \operatorname{Ch} x_1 \operatorname{Ch} x_2 + \operatorname{Sh} x_1 \operatorname{Sh} x_2$$

$$\operatorname{Th}(x_1 + x_2) = \frac{\operatorname{Th} x_1 + \operatorname{Th} x_2}{1 + \operatorname{Th} x_1 \operatorname{Th} x_2}$$

Double-argument formulas

$$\operatorname{Sh}(2x) = 2 \operatorname{Sh} x \operatorname{Ch} x$$

$$\operatorname{Ch}(2x) = \operatorname{Ch}^2 x + \operatorname{Sh}^2 x$$

Summary

Hyperbolic formulas:

- ▶ Addition: $\text{Sh}(x_1 + x_2) = \text{Sh } x_1 \text{ Ch } x_2 + \text{Sh } x_2 \text{ Ch } x_1$,
 $\text{Ch}(x_1 + x_2) = \text{Ch } x_1 \text{ Ch } x_2 + \text{Sh } x_1 \text{ Sh } x_2$
- ▶ $\text{Th}(x_1 + x_2) = \frac{\text{Th } x_1 + \text{Th } x_2}{1 + \text{Th } x_1 \text{ Th } x_2}$
- ▶ Double argument: $\text{Sh}(2x) = 2 \text{ Sh } x \text{ Ch } x$,
 $\text{Ch}(2x) = \text{Ch}^2 x + \text{Sh}^2 x$

Summary

Hyperbolic functions:

- ▶ Hyperbolic sine, cosine and tangent are defined through the exponential e^x
- ▶ Sh and Th are odd and increasing, Ch is even and positive
- ▶ Fundamental relation: $\text{Ch}^2 x - \text{Sh}^2 x = 1$

Formulas:

- ▶ Addition and double-argument formulas, analogous to the trigonometric ones but with different signs (for instance $\text{Ch}(2x) = \text{Ch}^2 x + \text{Sh}^2 x$)